English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, 19.01.2025 Test 2

The exact grading formula depends on the results. As an approximation, use 2 plus the points earned. Time allowed: 210 minutes. Good luck.

Problem 1. (1.5 points) Let ξ, η be independent random variables, $\xi \sim Exp(4)$ and $\eta \sim U(-1, 1)$.

1. (0.5 points) Find $\mathbb{E}[\xi\eta]$, $\mathbb{E}[\xi|\eta = 0]$, $\mathrm{Corr}(\xi, \eta)$ and $P(\xi \leq 0.25)$.
2. (0.5 points) Find the densities of $\xi^2 + 3$ and ξ/η.
3. (0.5 points) Let $Y_1, Y_2, Y_3, Y_4 \sim Exp(1)$ be independent. Let $Y_{(i)}$ be the ith smallest. Find the expectation and variance of $Y_{(3)}$.

Problem 2. (1 point) Two customers arrive simultaneously at a service system with two queues. If T is the service time,

$$T \sim \begin{cases} Exp(1), & \text{(slow queue) with probability 0.3,} \\ Exp(3), & \text{(fast queue) with probability 0.7.} \end{cases}$$

1. (0.5 points) Let T_i be customer i's service time. Find the distribution of $T_1 + T_2$.
2. (0.25 points) Find $E(T_1 + T_2)$.
3. (0.25 points) For large n, estimate the probability that the sum of n independent such service times exceeds $5n$.

Problem 3. (1 point) Let $X_1, X_2, \ldots$ be independent identically distributed random variables with density

$$f_X(x) = \begin{cases} 2x, & \text{if } 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

1. (0.5 points) Approximately how many observations n are needed for a 97% probability that the sample mean $\overline{X}_n := (X_1 + \cdots + X_n)/n$ lies between 0.5 and 0.7?
2. (0.5 points) Approximately what value do you expect for $\max_{1 \leq k \leq n} X_k$ when $n = 10\,000$?

Problem 4. (0.75 points) Let X, Y be independent, with $X \sim \mathcal{N}(2, 4)$ and $Y \sim \mathcal{N}(2, 9)$.

1. (0.25 points) Find their joint density and $P(X < Y)$.
2. (0.5 points) Calculate $E(|X - Y|)$ and $|E(X) - E(Y)|$. Can you determine which is larger without computing their exact values?

Problem 5. (1 point) Let ABC be a right triangle with legs $AC = BC = 1$. Through a uniformly chosen point P on AC, draw a line parallel to hypotenuse AB. Let Q, R be its intersections with the legs.

1. (0.25 points) Find the expected length of QR.
2. (0.5 points) Find the probability that two independently chosen random points inside ABC lie on opposite sides of QR.
3. (0.25 points) Let A, B be randomly chosen on a circle with centre O and radius 1. What is the expected area of AOB? What is the expectation of $AO \cdot AB \cdot \sin(OAB)/2$? Explain why the two quantities are or are not equal.

Problem 6. *(Bonus: 1 point)*

In classical (often called frequentist) statistics, to estimate an unknown parameter λ, we usually use methods such as maximum likelihood or the method of moments, without modelling λ itself as random. In the *Bayesian approach*, we instead assume a *prior* distribution for λ, such as $\Gamma(\alpha, \beta)$, and combine observed data with this prior knowledge to obtain its *posterior* distribution. We can then generate or predict new observations using a distribution that accounts for data already observed.

Models in which the posterior belongs to the same family as the prior, with updated parameters, are computationally convenient: the data have taught us parameter values that better describe reality. One such case is the exponential model $Exp(\lambda)$ with a gamma prior for the unknown λ: its posterior is also gamma.

Let $X_1, \ldots, X_n$ be independent observations from an exponential distribution,

$$X \mid \lambda \sim \mathrm{Exp}(\lambda),$$

with unknown parameter $\lambda > 0$. In the Bayesian model, assume the prior $\lambda \sim \Gamma(\alpha, \beta)$, with density

$$f_\lambda(x) \;=\; \frac{\beta^\alpha}{\Gamma(\alpha)}\, x^{\alpha-1}\, \exp(-\beta x), \quad x > 0.$$

1. (0.25 points) Prove that, after observing $X_1, \ldots, X_n$, the posterior distribution of λ is

$$\lambda \mid X_1, \ldots, X_n \sim \Gamma\Big(\alpha + n,\ \beta + \sum_{i=1}^{n} X_i\Big).$$

2. (0.25 points) Find the predictive distribution of a new observation X_{n+1} (i.e. the conditional probability $\mathbb{P}(X_{n+1} \in A | X_1, \ldots, X_n)$) and show that in certain cases it can be expressed using a *Pareto* random variable Y:

$$F_Y(x) = 1 - \left(\frac{a}{x}\right)^\alpha, \quad x > a,$$

for a parameter a. This is a classical example of a heavy-tailed distribution. (*You may integrate with respect to x the distribution* $\mathbb{P}(X_{n+1} \in A | \lambda = x) \times f_{\lambda | X_1, \ldots, X_n}(x | X_1, \ldots, X_n)$.)

3. (0.5 points) Find $\mathrm{E}(\lambda | X_1, \ldots, X_n)$ and $\mathrm{Var}(\lambda | X_1, \ldots, X_n)$. Recall that the conditional variance of Y given X is defined as

$$D(Y|X) \;=\; E\big[(Y - E(Y|X))^2 \,\big|\, X\big].$$

Use this definition also to prove that

$$D(Y) \;=\; E\big[D(Y|X)\big] \;+\; D\big(E[Y|X]\big).$$

4. *(Bonus to the bonus)* How would you interpret the difference between a classical confidence interval and a Bayesian highest-posterior-density interval for λ in this example?